

T SIVANI

SOFTWARE ENGINEER · FULL-STACK & AI SYSTEMS

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PROFESSIONAL SUMMARY

Full-stack Software Engineer who works across frontend, backend, and databases: production experience shipping cross-platform web and mobile interfaces integrated with **REST and GraphQL APIs**, plus **Python (Flask) and PostgreSQL** backends built end-to-end, from system architecture and relational schema design to automated testing and deployment. Hands-on with the systems underneath AI applications, having built a **vector search engine** of the kind that powers vector databases and an **ML monitoring platform** for drift detection. Holds the IBM AI Developer Professional Certificate.

TECHNICAL SKILLS

Frontend:	React.js, Next.js, TypeScript, JavaScript, HTML5/CSS3, Flutter, Dart, Figma
Backend & APIs:	Python (Flask), RESTful APIs, GraphQL, Node.js, Express, Microservices
Databases:	SQL (PostgreSQL, MySQL, PL/SQL), Relational Schema Design
AI / ML Systems:	Prompt Engineering, Structured Outputs, Tool Calling, Vector Search (HNSW, ANN), ML Model Monitoring, Drift Detection (PSI, Kolmogorov-Smirnov), AI Application Development
Architecture & Tools:	System Design, Distributed Systems, Git, GitHub, Pull Request Reviews, Pytest, Linux/Bash, Full-Stack Deployment

WORK EXPERIENCE

Software / Front-End Developer | Coreillustrio 06/2025 – 08/2026
Chennai - Guduvancherry, India

- Engineered **cross-platform web and mobile interfaces** using Flutter/Dart and React concepts, translating Figma mockups into responsive, pixel-perfect user experiences across iOS, Android, and web.
- Integrated **REST and GraphQL APIs** into frontend state architecture, optimizing application rendering speed, network data fetching, and mobile application performance.
- Shipped rapid feature iterations in agile sprints through Git pull-request workflows, conducting code reviews for teammates.

Data Analyst Intern | Larsen & Toubro 06/2023 – 07/2023
Chennai - Guindy, India

- Designed and normalized **relational database schemas** using MySQL Workbench, resolving data inconsistencies and optimizing query structure for enterprise datasets.
- Presented database normalization concepts and relational data modeling solutions to engineering leadership.

FULL-STACK & AI SYSTEMS PROJECTS

ModelWatch: ML Drift Detection Platform | Python, Flask, PostgreSQL, SQL 2026

- Engineered a **full-stack platform** that simulates production ML traffic and detects real-time feature drift using Population Stability Index (PSI) and Kolmogorov-Smirnov tests implemented from first principles.
- Designed a **PostgreSQL schema** tracking runs and prediction logs over time, with automated triggers for model re-training when configurable drift thresholds are breached.

HNSW Vector Search Engine | Python, Vector Search, Data Structures 2026

- Implemented the Hierarchical Navigable Small World (HNSW) algorithm from first principles: the **approximate nearest-neighbor index that powers vector databases** and retrieval pipelines.
- Benchmarked against a brute-force baseline across **50K vectors**, achieving significant query speedups while maintaining high recall@10 accuracy.

Raft Consensus Distributed Systems Simulator | Python, Distributed Systems, Concurrency 2026

- Implemented the Raft consensus algorithm from scratch, simulating a multi-node cluster with leader election, log replication, and majority-based commits on a deterministic clock.
- Authored **automated test suites** verifying split-brain prevention during concurrent elections and automatic leader recovery after simulated node failures.

Repo Doctor CLI: Git Static Analysis Tool | Python, Pytest, Static Analysis, Git 2026

- Built a pip-installable CLI that runs pattern-based static analysis over Git repositories to flag leaked secrets, oversized files, and missing configuration files, with color-coded terminal reporting.
- Shipped with a **full pytest suite** that validates every check against fixture test repositories.

WAL + MVCC Transaction Engine | Python, SQL, Storage Engines, System Design 2026

- Built a transactional key-value store with Write-Ahead Logging (WAL) for fault-tolerant crash recovery and MVCC for snapshot-isolated reads with write-write conflict detection.

EDUCATION

B.Tech in Computer Science and Engineering (CGPA: 7.38) 01/2021 – 12/2025
SRM Institute Of Science and Technology | Kattankulathur, Tamil Nadu

CERTIFICATIONS

- IBM AI Developer Professional Certificate – Coursera
- Building AI Powered Chatbots Without Programming – Coursera
- Introduction to AI – Coursera
- Developing AI Applications with Python & Flask – Coursera
- Python for Data Science, AI Development – AWS